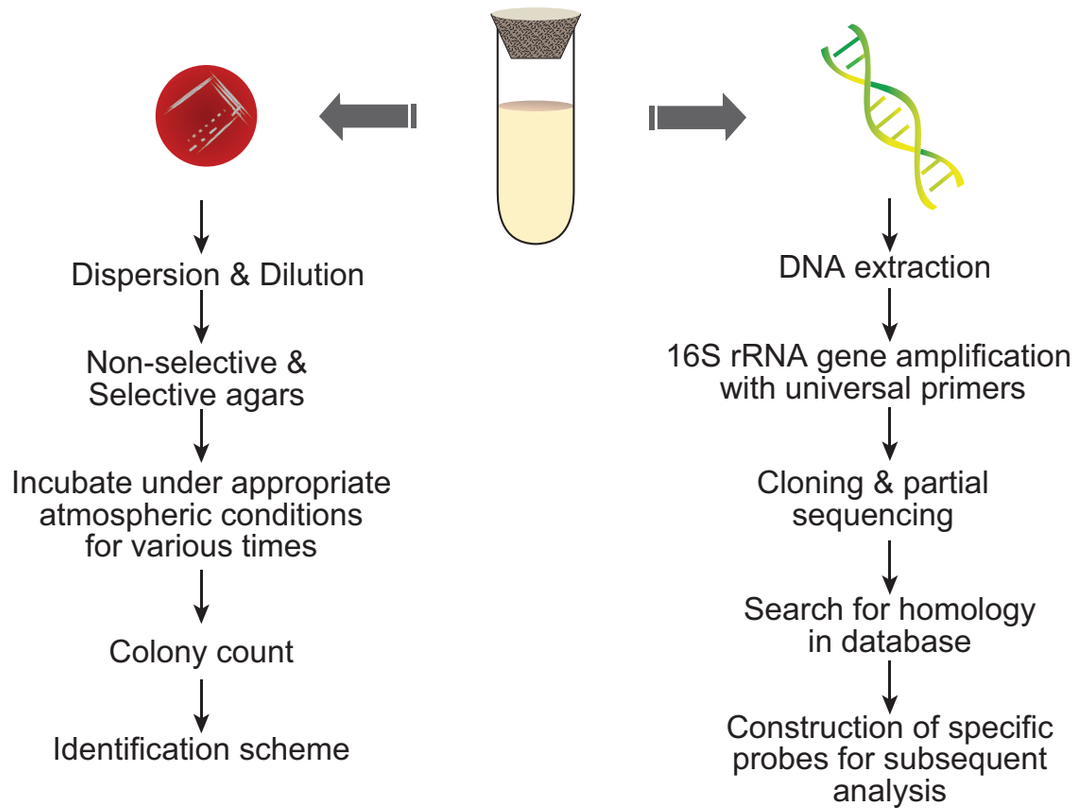


Figure 8.4: General scheme for classification and identification in microbial taxonomy

groups. Numerical taxonomy also yields classification that has a high degree of stability and predictability.



In Numerical Taxonomy, which was defined by Peter H.A.Sneath and Robert Sokal, each characteristic is given equal weightage and it is converted into numerical form and compared by means of a computer.

Atleast 50 and preferably several hundred characterictics are compared.

The presence and absence of selected characters in the group of organism is calculated by simple matching coefficient (SSM), called Jaccard coefficient.

Molecular Taxonomy



Molecular techniques in the field of biology has helped to understand genetic relationship between the numbers of different taxonomic categories.

DNA and protein sequencing, immunological methods, DNA-DNA or DNA-RNA hybridization methods are very helpful in studying different species.

The data or information from such studies are used to construct phylogenetic tree (a branching diagram showing the evolutionary relationship among various biological species based on similarities and difference in their physical or genetic characteristics).

TN_GOVT_XI_Micro_Biology_CH08.indd 116 16-02-2019 10.01.27 AM